

Review

The training process: Planning for strength–power training in track and field. Part 1: Theoretical aspects

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Abstract

The process of strength–power training and the subsequent adaptation is a multi-factorial process. These factors range from the genetics and morphological characteristics of the athlete to how a coach selects, orders, and doses exercises and loading patterns. Consequently, adaptation from these training factors may largely relate to the mode of delivery, in other words, programming tactics. There is strong evidence that the manner and phases in which training is presented to the athlete can make a profound difference in performance outcome. This discussion deals primarily with block periodization concepts and associated methods of programming for strength–power training within track and field.

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1. Introduction

There is little doubt that the method employed makes a significant difference in the physiological and performance adaptations resulting from a resistance-training program.¹ For example, high-volume programs generally have a greater influence on body composition, enhancement of muscle cross-sectional area, and work capacity factors than do low-volume higher intensity programs. In contrast, high-intensity programs have a greater influence on maximum strength, peak power, and velocity compared to low-intensity programs.^{2,3} Evidence also indicates that the level of the athlete results in somewhat different adaptations to training at least quantitatively. For example, among relatively untrained or weak athletes, strength training will provide as great or greater increases in power as compared to power training alone. Furthermore, evidence indicates that prior strength training or having higher maximum strength levels can potentiate further power training.^{3–7}

Training should be recognized as a process which prepares an athlete, technically, tactically, psychologically, physiologically, and physically for the highest possible levels of performance.⁸ First is the realization that training is a process

that requires considerable forethought and planning. Training is multi-factorial in nature; as such the attempt is to exploit known principles of physics, physiology, and psychology in order to maximize the effects of the training stimulus. The training process is vitally concerned with the positive enhancement of performance—therefore the process must provide:

- an appropriate stimulus for adaptation;
- an appropriate means for assessing progress (monitoring);
- and additional means beyond sets and reps (i.e., stimulus) including rest–recovery phases, psychological re-enforcement, daily nutrition, supplements, sleep, *etc.* so that recovery–adaptation is optimized.

The sport training process attempts to take the athletes as close as possible to their genetic limits, thus training is not simply recreational exercise. Considering this concept, a good coach should be viewed in the same context as a good physician. Therefore the training process can be viewed as a prescription.

When developing a training process there are several “realizations” that are major factors in the development of a successful training process:

Every athlete does not progress to the elite level.⁹ There is no substitute for innate talent (genetics). There are two aspects to this realization. First, there are genetically linked physiological characteristics that relate to superior performance. These heritable characteristics range from higher testosterone

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concentrations in both men and women to differences in muscle fiber types.^{10–12} As a result of these genetic links athletes with specific traits simply are able to perform better in specific sports. For example: athletes with more Type I motor units typically have a higher $\text{VO}_{2\text{max}}$, handle lactate more efficiently and have endurance event advantages.¹³ Second is the relationship between heredity and the training “window of adaptation”.^{10,14,15} Everyone responds to a well-planned training stimulus; however, because of heredity some athletes respond to the same program with greater adaptation, and so are able to progress further than typical athletes. Athletes that have both traits are most likely to progress to the elite level.

Being stronger can make a substantial difference in performance. Maximum strength is related to:^{4–6,8,16–20}

- Magnitude of force production with greater peak and average forces that may allow for higher velocities and power outputs to be achieved when using submaximal loads;
- Rate of force development (RFD) which can describe faster muscle activation with resultant greater force during critical time periods;
- Superior ability to develop and respond to stretch shortening cycles (SSC);
- Greater peak and average power, which means that work is accomplished at a higher rate;
- Greater absolute endurance, especially high intensity exercise endurance (HIEE) that may reduce “central” (nervous system) fatigue; i.e., more total work can be accomplished;
- Greater postural strength, which describes the ability to hold static and dynamic positions better during performance;
- Some evidence that force sensitivity and sensation is superior (may be a result of strength training partially independent of maximum strength)—ability to appropriately modulate force production is superior;
- Importantly, among weaker athletes (most scholastic and collegiate athletes), strength training alone may produce equal or superior results in terms of RFD, power, *etc.* compared to power or speed training,^{4–6,8} particularly when performed in an integrated fashion with sport training programs.

2. Components of the training process

A major component of planning concerns the understanding of both the developmental and conceptual aspects of the training process. Considering that training is multi-faceted, the degree of success achieved will relate to how well a coach learns to plan conceptually and subsequently creates detailed subcomponents oriented to produce superior results. These factors of the training process include the following.

2.1. Construction of the annual plan⁸

The most important aspect of the planning process is the creation of a sound annual plan. The annual plan is a road map for the overall training process and its development is the first step in constructing the athlete’s program (the program can be found online at [doi:10.1016/j.jshs.2015.07.003](https://doi.org/10.1016/j.jshs.2015.07.003) as Table S1). The annual plan lists the coming competitions, the projected

testing/monitoring dates and a general guide to conceptual training (periodization and programming).

Conceptually, periodization is a blueprint dealing with fitness phases and timelines. This blueprint allows the coach to project an approximate timeline for when various types of fitness phases (e.g., endurance, strength–endurance, strength, power, speed, taper, *etc.*) will be emphasized, the order of the phases and how long each of these phases will last. The annual plan also contains the programming for the periodized training plan.

Programming deals with creation of appropriate sets, repetitions and exercise selection that make the different phases efficient and efficacious. Programming is a primary factor in fatigue management and directing training toward the desired goal(s). Excessive accumulative fatigue inhibits physiological adaptation to training stimuli, produces non-beneficial psychological aspects and increases the injury potential. Thus a primary aspect of programming deals with appropriate variation of training volume, intensity, and exercise selection such that fatigue is controlled and adaptation is optimized.

The separation of these two aspects (periodization and programming) within the training process is often difficult. Depending upon the type of periodization process chosen (e.g., classic *vs.* block), the programming can be different. Regardless, development of the annual plan is the primary step in creating an efficient and successful training process.

Most important, the success of a training plan resides in how collaborative and willing the head coach and assistant coaches (and strength–conditioning coaches) are to learn the intricacies of, and become fully immersed into, the planning process, a factor that does not always occur. Predictably, for coaches not accustomed to a conceptual planning process, a difficult, but temporary, adjustment period often takes place.

2.2. Factors during the annual plan development

A sound understanding of the basic training principles and their application during training can make a substantial difference in training process outcomes.^{1,7,8,21–24} There are four basic training principles: overload, variation, specificity, and reversibility. When these principles are appropriately addressed and properly integrated into the training process as a result of logically applied programming, adaptation is optimized, fatigue management is enhanced, overtraining potential is reduced, and the potential for superior performance is augmented.

2.2.1. Overload

Overload entails providing an appropriate stimulus for attaining a desired level of physical, physiological, psychological, and performance adaptations. Overload can be conceptualized as a training stimulus that forces the athlete beyond normal levels of physical performance. Thus the application of an appropriate overload stimulus can include range of motion, absolute and relative intensity (RI) levels, frequency, and time factors. All overload stimuli will have some level of intensity, RI (percentage of one repetition maximum, 1 RM), and volume. The determination of appropriate overload stimuli for different modes (weights, variable resistance devices, semi-isokinetic devices, *etc.*) of resistance training can be challenging. Quantification of

some forms of overload or some forms of resistance, for example elastic bands, is at best difficult, particularly as the bands age. For this discussion, the quantification of overload stimuli will deal with weight training.

2.2.2. Variation

Variation describes the removal of linearity from the training plan by manipulating the overload characteristics and degree of specificity. This factor is important for the production of strength–power characteristics and producing a power spectrum. In addition, varying the load may be the most important factor in guiding training toward a specific goal while accounting for “fatigue management”. Appropriate variation is an essential consideration for continued adaptation over a long-term training program.^{8,25} Variation involves manipulation of the training intensity, speed of movement, volume, rest and recovery periods, and exercise selection. The enhancement of a number of abilities and skills may occur through the emphasis and timeliness of the aforementioned variables in appropriate sequences.⁷

Indeed appropriate sequencing and emphasis of fitness characteristics are part of the foundation of “block periodization”. Consideration of several different levels of variation is necessary in a training program (i.e., quadrennial, annual, intermediate, weekly, day-to-day, *etc.*). The level of variation in the training program is directly related to the level of the athlete, with advanced athletes requiring a greater degree of variation as compared to novices and beginners.

2.2.3. Specificity

Specificity is the degree of metabolic and mechanical similarity between the desired performance characteristics and training exercises. The aim of increased specificity throughout the training process is to enhance the transfer of training effect, which is a measure of how much the training transfers to actual sport performance. If athletic performance enhancement is a primary goal, then specificity of exercise and training arguably becomes the most important consideration for selecting methods and modes for resistance training. There are two basic aspects to specificity: mechanical and metabolic.

Mechanical specificity refers to the kinetic (force) and kinematic (displacement, velocity, power, RFD) associations between a training exercise and a physical performance. Mechanical specificity includes movement patterns (e.g., range of movement, open vs. closed kinetic chain exercise, *etc.*), peak force, rate of force development, acceleration patterns, and velocity parameters. Conceptually, mechanical specificity has strong support based on observations of inter- and intra-muscular task specificities.^{1,8,26,27}

Task specificity is concerned, at least in large part, with the manner in which the motor cortex organizes motor unit activation (intra-muscular tasks) and whole muscle activation patterns (inter-muscular tasks). There is good evidence that both intra- and inter-muscular task specificity aspects play a major role in strength–power training. Basically, the greater similarity training exercise has to the actual physical performance, the greater the probability of transfer.^{1,24,28,29} For example, intra-muscular task specificity suggests that for a specific activity

only a defined motor unit task group will be activated.^{29,30} Evidence of this can be found from practical and research aspects. For example, bodybuilders have noted that to fully develop a muscle, a variety of different exercises for that muscle must be performed; an observation that is supported by considerable research. Several studies^{26,27,31–33} have indicated that as a result of resistance training muscle hypertrophy does not occur uniformly throughout a muscle nor does it occur uniformly in different regions (e.g., upper vs. lower body). This brings up the possibility of indiscriminant hypertrophy, which could interfere with performance. For example, Abe et al.²⁶ noted that among sprinters most of the hypertrophy as a result of training was relatively localized in the proximal (upper) portion of the thigh, which tends to reduce the moment of inertia for the hip joint, providing an advantage for sprinting. Thus, training which emphasizes hypertrophy of the distal portion (lower) of the quadriceps may not be as advantageous.

2.2.4. Reversibility

Reversibility describes the loss of fitness that originally accumulated as a result of the training stimulus. This reversal of fitness can occur as a result of two factors.

First, a removal or reduction of the stimuli can result in de-training, which can be considered a “negative” physiological adaptation. For example, muscle atrophy may ensue as the result of resistance training being removed or through a haphazard and marked reduction in volume.

Second, involution of performance characteristics, which can be described as a diminished performance capability, occurs even though the stimulus is still being applied. This type of reversibility often arises in relation to a monotonous, relatively unchanging, program that is likely a result of too little mechanical variation in the training program.

A type of involution can also occur as a result of maladaptations resulting from poor fatigue management (i.e., non-functional overreaching). Detraining is most common after any substantial decrease in volume. Through observation and discussion with athletes and coaches world-wide, involution is not uncommon after 12–16 weeks of training the same basic fitness characteristic (i.e., basic strength, power) even though there can be considerable variation in the exercise type and, to an extent, sets and repetitions.

2.3. Additional training considerations

While the planning process is guided by the four basic training principles, additional factors such as intensity, training volume, training sessions, and training density assist in how well the planning process addresses the tenants of periodization.

2.3.1. Intensity

Intensity is often misunderstood as a component of within-exercise prescription. There are several aspects to the intensity component. Intensity factors are associated with the velocity of movement, rate of performing work, and the rate at which energy is expended.² Intensity factors can be separated into two aspects: training intensity (TI) and exercise intensity.^{8,34}

TI is concerned with the rate at which a training exercise or training session proceeds; it can be estimated by the average load lifted per exercise, per day, per week, *etc.*, and relates to the training density. For example, within a session, the average load lifted is directly related to the time taken to complete the exercise; thus, for a specified number of repetitions, a heavier load requires more time to complete. Furthermore, a greater number of repetitions lead to a longer duration of the set. Considering these factors, when normal self-selected rest periods ($\approx 2\text{--}5$ min between sets) are utilized, the work-rate can be dictated by loading and repetition number (Table 1).

The RI is a percent of the 1 RM. The 1 RM value is relatively stable only for advanced strength-power athletes. Thus, using RI to plan resistance-training programs must be carried out with this aspect in mind. Exercise intensity is the actual power output of a movement.

2.3.2. Training volume

Training volume is a measure of the total work performed during training and is strongly related to total energy expenditure and subsequent metabolic alterations.^{35–39} The resistance training volume is a function of the load, number of repetitions and sets per exercise, the number and types of exercises performed (e.g., large vs. small muscle mass), rest periods and the frequency (i.e., number of times per day, week, month, *etc.*) with which these exercises are repeated.

Volume load (VL) is a more reasonable estimate of the amount of work accomplished during training and is commonly used in both research and practical settings. VL is calculated by summing the product of the load and the number of repetitions for each set. This should be done for each exercise within the desired time frame (e.g., weekly, monthly, *etc.*) and the sum total for the combined exercises represents a reasonable

estimate of the total work accomplished. However, this approach may not work well if major changes in the exercises performed occur, or if the displacements of the exercises are altered substantially. For example, partial movements replacing full movements (i.e., 1/4 squats vs. full squats) during the competition phase. In order to then make better comparisons, the VL for each exercise should be multiplied times the vertical distance the bar moves.⁴⁰ The vertical displacement can easily be measured (estimated) with a steel tape measure.

2.3.3. Training session

The application of TI and training volume can be considered in terms of the training session (i.e., all of the exercises performed during a specific period) or in terms of single exercises. An understanding of overload factors can aid in the programming of training, including methods (i.e., sets and repetitions), velocity of exercise, and exercise selection. The interaction/association of VL and TI can be illustrated by calculating these factors for two sample-training sessions (using the squat as an example). Table 1 displays data for training Day 1 (the general preparation (GP) phase) and Day 2 (the late preparation phase). In this example, the VL for Day 1 was larger than that for Day 2 (7000 vs. 4700 kg); however, the TI was larger for Day 2 than for Day 1 (134 vs. 116). VL and TI are inversely related. Furthermore, TI is directly related to the rate at which the load is lifted (kg/s) and is an indication of the rate of training. Calculation of the TI, while reflecting work rate (kg/s), is less time-consuming than measuring and calculating kilograms per second and thus has a practical advantage. The average VL and TI can be easily calculated per week, month, or phase of training. In this manner, using these variables, a reasonable record of training progress can be made.

Table 1
Session 1: general preparation (accumulation) and late preparation/pre-season (accumulation).

	Set	Repetition	Load (kg)	VL	TI	Duration of set (s)	Work rate (kg/s)
General preparation (accumulation)							
	1	10	60.0	600.0		30.0	20.0
	2	10	100.0	1000.0		35.0	28.6
	3	10	120.0	1200.0		40.0	30.0
	4	10	140.0	1400.0		45.0	31.1
	5	10	140.0	1400.0		47.0	28.8
	6	10	140.0	1400.0		48.0	29.2
Total	6	60	700.0	7000.0		245.0	167.7
Mean/set		10	116.7	1167.0	116.7	40.8	28.0
Late preparation/pre-season (accumulation)							
	1	5	60.0	300.0		15.0	20.0
	2	5	100.0	500.0		17.5	28.6
	3	5	140.0	700.0		20.0	35.0
	4	5	180.0	900.0		25.0	36.0
	5	5	180.0	900.0		26.0	34.6
	6	5	180.0	900.0		27.0	33.3
	7	5	100.0	500.0		15.0	33.3
Total	7	35	940.0	4700.0		145.5	220.8
Mean/set		5	134.3	671.4		20.8	31.5
Mean/set not including down set		5	134.2	700.0	134.3	21.8	32.1

Abbreviations: VL = volume load, sum of load lifted (repetitions $\times$ mass); TI = VL/repetitions (or mean load/set).

Based on Refs. 8 and 24.

2.3.4. Training density (TD)

TD is related to the volume of training and deals with the frequency of training per session, per day, per week, *etc.* TD can also be manipulated to allow higher training intensities to be maintained. For example, training Day 1 might contain three training sessions, each of equal volume while training Day 2 might contain one training session of similar volume to Day 1. Day 1 would have a higher TD and the average intensities of each smaller volume session could be maintained at higher intensities compared to Day 2. One week could consist of six training sessions, while another could contain 12; the TD in Week 2 would be higher. TD can be manipulated to alter planned, functional overreaching and tapering paradigms.⁴¹ For example, TD could be higher in the potentiating aspects of planned over-reaching (high volume phase).

3. Understanding the basic periodization concepts(s)

Periodization provides the overall concept of training and deals with subdividing the training process into specific time periods and fitness phases.^{8,42} DeWeese et al.⁴³ have recently reviewed and discussed various definitions of periodization. One important factor present in the various definitions presented by DeWeese et al.⁴³ is that fitness phases, inherent in the periodization concept, occur in a cyclical in manner. Indeed the periodization concept and subsequent programming (alterations in exercise selection, volume, and intensity) are designed to remove linearity. It should be noted here that the recently coined term “linear periodization” is a misnomer both by definition and by concept. Periodization is not linear but rather exploits the basic training principles, particularly variation. Furthermore, general periodization terminology is not typically agreed upon and varies from country to country and from author to author.⁸ As a result, the following will be used for this discussion. Periodization is the logical, sequential, phasic method of manipulating training variables in order to increase the potential for achieving specific performance goals while minimizing the potential for overtraining and injury through the incorporation of planned recovery.^{8,25,43} While terms and the phrasing of time periods differ slightly as a result of programming theories,^{8,44} Table 2 describes the various periods of time that are commonly used within the periodization process.

In addition to the time periods described above, periodization calls for the segmentation of the training plan into dedicated fitness phases. In other words, periodization considers that phases of fitness development will move from general to more specific training as the athlete nears competition. These fitness phases are described in Table 3.

3.1. Programming considerations

As a result of incompatibility with the modern training schedule, classic or traditional periodization (TP) may not produce desired effects.^{8,45} Current evidence indicates that phase potentiation strategies that are inherent in Block periodization produce superior results, particularly as it relates

Table 2
Timelines for the planning process.

Term	Timeframe
Annual plan	Approximately a year of planned training.
Macrocycle	Several months to approximately 1 year. Consists of five phases related to fitness emphases: (1) general preparation, (2) special transition (1st transition), (3) competition, (4) taper, (5) active rest (2nd transition). This period of time typically contains summated mesocycles.
Mesocycle	The addition of summated blocks where several weeks (typically 4–16 weeks) align to allow for the accumulation, transmutation, and realization from a training stimulus.
Summated microcycles	The linkage of single weeks in which the volume and intensity of training are manipulated in a manner leading toward achieving a specific goal. (e.g., 3 weeks of increasing volume followed by an unload week). A summated microcycle can also be termed a “block” which associates with a concentrated load in which one aspect of fitness is being emphasized while all others aspects are de-emphasized.
Microcycle	A period of several days, also known as summated sessions, that share a similar theme or goal.
Session	A single training practice or workout.

to athletes (Table 4). Indeed there are several problems inherent in the traditional approach to periodization that may in part or whole be obviated by the block method.

- The modern competition schedule is not in concert with TP timelines. Potentially an athlete cannot hold a true peak for more than 3 weeks,^{46–49} which might work well if there is only one major competition for each macrocycle; however, there may be multiple important competitions, sometimes relatively close together. Thus, the TP approach that contains only a limited number of peaks likely will not result in adequate outcomes.
- Many coaches (not understanding all of the subtleties of periodization) attempt to increase the volume of many different training variables simultaneously. This approach can cause difficulty in fatigue management due to high volumes of training as well as create physiological adaptations, which do not favor the goal of the periodization phase. It is very difficult and potentially less productive to attempt the simultaneous development of different physical and physiological characteristics or motor abilities.^{8,25,49,50} This loss of productivity can occur for several reasons such as: (A) simultaneous training of fitness characteristics typically result in a substantial increase in training volume that can increase recovery time;^{28,36,51,52} (B) high volume or mixed training methods combining endurance and strength–power related training can reduce effectiveness of training direction (i.e., achieving specific goals) and stagnate gains in performance and usually favor endurance over strength and power.^{52–59} Thus, the goals of the training program could be compromised. One aspect that can compound the problem associated with mixed training methods (i.e., simultaneously targeting multiple skill and fitness variables) is consistent training to failure or near failure (i.e., high absolute or relative intensities)—a problem that is particularly prevalent for resistance training.

Table 3
Fitness phases.

Term	Timeframe
GP	Typically denotes a higher volume, low intensity, relatively low mechanical specificity phase with the intention of raising sport specific fitness, which includes altering body composition and raising work capacity.
Special preparation	A relatively high volume, low to moderate intensity, higher mechanical specificity oriented phase. A portion of this phase typically emphasizes the athletes' ability to repeat exercise with a greater mechanical specificity. This phase can be used to transition from higher volume, less-specific GP training to a higher-intensity, very specific training phase that closely associates with competition.
Competition phase	Commonly refers to a moderate to low volume phase with moderate to high intensity that is mechanically specific. The purpose is to maintain fitness while enhancing technical consistency/efficiency. The competition phase can last several months and may contain periods of mini-preparation, if volume reduction should last more than 12–16 weeks. These mini-preparation periods may include periods of functional overreaching which appear to enhance competition performance if timed correctly.
Peaking phase	A segment of the competition phase lasting a short time (usually ≤ 4 weeks), that takes place just before major competitions. During the early portion of peaking, volume can be reduced and training intensity (or exercise intensity, depending on the sport or performance goals) is increased or maintained at relatively high levels. Typically, during the last few days before important competitions, intensity factors are also reduced to encourage adequate recovery. The later portion of a peaking phase (typically 8–14 days), when volume is markedly reduced, is referred to as a "taper". ^{8,46,47} A taper consists of a reduction of training volume in order to reduce fatigue and take advantage of the fitness–fatigue paradigm. ^{8,25,46,47} A taper can be coupled with planned overreaching in order to boost performance beyond that of a typical taper.
AR	A period of recovery after peaking and major competitions (usually 1–2 weeks) using a reduce volume and intensity of training). Recovery includes healing and rehabilitation of any injuries that may have occurred as well as recovery from the emotional rigors of competition. Complete rest allows sports-specific fitness to deteriorate to a degree from which it is difficult to recover without extensive training. Compared to complete rest, AR allows for less deterioration of fitness and a faster return to peak fitness during the next cycle. AR usually lasts about 1 week. At times the training may be re-directed into another activity to improve the psychological/emotional recovery aspects (make the training more recreational).

Abbreviations: GP = general preparation; AR = active rest.

Some of these problems associated with TP were recognized by Matveyev⁴² and Matveyev and Zdorniy⁵⁹ as early as the late 1960s. However, training paradigms that addressed these problems were not well formulated until the 1970s and 1980s. The basic concept of Block periodization was formalized during this time period by Verkoshansky,^{60,61} Issurin and Sahrobajko,⁵⁵ and Issurin⁶² in Europe (Soviet Union) and by Stone and colleagues^{44,63} in the US, particularly for strength–power training.

3.2. Block programming

Block periodization depends upon several observations and characteristics.^{8,44,50,62,63} This begins with the observation that the development of specific physiological/performance characteristics at high levels cannot be sustained for long periods of time. In conjunction, the attempt to simultaneously develop multiple characteristics is often counterproductive. As a result, emphasizing training variables at different points of the training year can develop specific performance characteristics. This tenant is met through the utilization of a series of "concentrated loads" that are sequenced together at the appropriate time in order to produce superior results. In addition, "planned functional overreaching" can promote additional adaptation while taking advantage of the fitness–fatigue paradigm. Collectively, the summing effects of concentrated loads coupled alongside planned functional overreaching strategies may positively affect future training through phase potentiation, as a result of the accumulation of residual effects.

A comparison of block and TP is shown in Table 4. These differences highlight the conceptual advantages of block periodization methods. Coaches should understand that the differences between winning and losing, a medal and no medal, are often extremely small. In the last eight Olympics the

differences between 1st (Gold) and 4th (no medal) has been less than 1.5% for most sports and events. Thus training advantages, even those seemingly trivial, may in actuality be quite large in the realm of competitive sport. There is little doubt that block periodization concepts and methods can offer training advantages.

4. Factors affecting the training process outcome

There are a number of factors that will positively or negatively affect training outcomes; these include diet, supplements, sleep, *etc.* This discussion will center on those directly related to training.

First is the realization that rapid gains are not always in the overall best interest of the athlete. Fig. 1 shows the relationship between the intensity of training and performance gains. Generally, the higher the relative intensity, the faster the gains occur; however, this takes place by sacrificing the final performance level and the length of time a high level of performance can be sustained. Thus, in settings in which athletes may take time off (e.g., summer, Christmas break, *etc.*) very intense (or very high volume) programs designed to get the athlete back into shape rapidly may ultimately decrease the level of attainable performance. Training programs are likely to produce rapid gains in performance, early performance plateaus, and rapid diminished returns including:^{8,41,64–68}

- Constant high absolute or relative intensity (e.g., Bulgarian System in weight-training)
- Weight-training use of RM zones and subsequent training to failure on a regular basis;
- Linear programs using consistent increases in intensity.

Next is the ability of the athlete to make maximum efforts when necessary. Fig. 2 illustrates the relationship between

Table 4
Block periodization studies among athletes (preserved ecological validity).

Study	Athlete	Length	Comparison	Major finding	Comments
Issurin and Sahrobajko, ⁵⁵ Issurin et al. ⁷⁴	23 male canoe/kayak athletes	3 years (seasons)	TP vs. BP	BP > TP on water power output, propulsion efficiency, ergometer power; Superior performance in international events	BP group superior performance in world level events
Harris et al. ⁷	42 American football players HF = 13 HVP = 16 BP (COM) = 13	9 weeks	BP (COM) vs. HF and HVP	BP improved to greater extent and on wider variety of measures than other two groups	All groups preceded study using 4 weeks SE CL; Planned comparison: COM (BP—only group finishing block); Winter workouts (sprints, jumps, <i>etc.</i>)
Garcia-Pallares et al. ⁵⁴	World class male kayak	2 years (seasons)	TP vs. BP	BP > TP VO _{2peak} (11% and 8.1%) BP > TP PS (peak), Pw (peak), and SR (peak)	BP showed greater improvements even though TP cycle was 10 weeks and 120 training hours longer than the BP cycle
Breil et al. ⁵³	21 alpine skiers (junior elite) BP = 13 TP = 8	11 days	TP vs. BP	BP > TP VO _{2peak} , cycle PP, and power at Ven Threshold	
Mallo ⁷⁵	77 male professional soccer players	4 years (seasons)	BP vs. T soccer training	BP performed (competition and measures, e.g., 1 RM, speed endurance, <i>etc.</i>) better than traditional	Each season divided into three complete blocks; Team performance was assessed in competition, by percentage of points by team in each match examined in relation to the training phase
Painter et al. ⁷⁶	26 day = 1 track and field strength power BP = 12 DUP = 14	10 weeks resistance training with normal training	BP vs. DUP	Based on ES and % change BP > DUP in 1 RM squat, IPF, and RFD; BP > DUP work efficiency BP > VO _{2peak} , time to exhaustion	Groups equated initially on 1 RM, event and sex; DUP performed approximately two times more work
Bakken ⁷⁷	World class X-country and biathlon	5 weeks intensive endurance training	BP vs. T training		
Rønnestad et al. ⁵⁸	15 male cyclists BP = 8 TP = 7	12 weeks preparation training	TP vs. BP	BP > TP: VO _{2max} , 2 mmol/L La, and during 40 min trial and >Hg mass	Similar V and I
Rønnestad et al. ⁷⁸	19 male cyclists BP = 10 TP = 9	4 weeks endurance training	TP vs. BP	BP > TP: VO _{2max} , 2 mmol/L La, and PP (Wingate) TP did not change	Similar V and I
Bartolomei et al. ⁷⁹	24 strength power athletes (track and field, rugby) BP = 12 TP = 12	15 weeks resistance training	TP vs. BP	BP > TP Upper body strength measures (impulse optimal PP, 1 RM)	Changes based on ES—also possible to conceptualize as block vs. block as TP was actually divided into two summated microcycles;
Rønnestad et al. ⁸⁰	19 elite cross-country skiers BP = 10 T (TP) = 9	5 weeks endurance training	BP vs. TP	BP > TP Larger peak power output and power output at blood lactate concentration of 4 mmol/L Maximal oxygen uptake increased by 2% ± 2%	Similar V and I

Notes: These 11 studies show and discuss comparisons of block periodization protocols vs. other types of training including traditional periodization methods of training. They are quite unique in that the studies all used athletes. These studies include a variety of sports, training characteristics (e.g., strength, endurance, *etc.*) and a variety of timelines from very short (11 days) to 4 years (seasons). In every case the block model (BP) showed evidence of superior adaptation. Preserved ecological validity—ecological validity deals with studying groups in their natural or typical cultural setting so that generalization to other similar groups has a greater validity. Thus studies with athletes in their normal training environment preserve ecological validity.

Based on Issurin.⁶²

Abbreviations: BP = block periodization; TP = traditional periodization; HF = high force group; HVP = high velocity high power group; COM = combination group; SE = strength endurance; CL = concentrated load; DUP = daily undulating periodization; ES = effect size; IPF = isometric peak force; RFD = rate of force development; PS = paddling speed; Pw = power output at VO_{2peak}; SR = stroke rate; PP = peak power; La = lactate; T = traditional; V = volume; I = intensity.

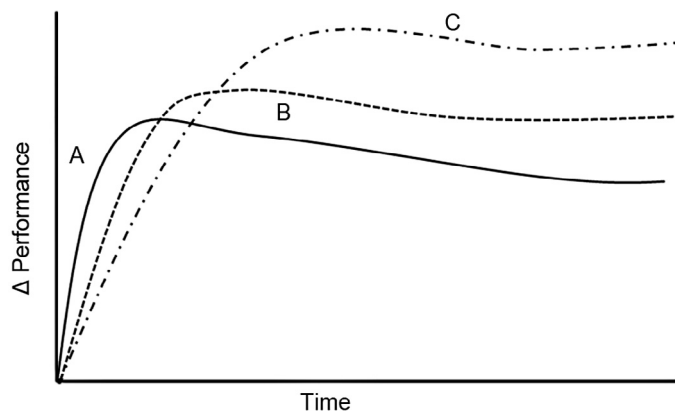


Fig. 1. Characteristics of different prolonged average intensities. A: Rate of gain is directly related to average intensity of training; B: final level of performance is inversely related to the average rate of gain; C: rate of gain is inversely related to the length of time (time period) that peak performance can be sustained. Thus for: 1. initial rate of gain $A > B > C$; 2. final level of performance $C > B > A$; 3. time period of maximum performance $C > B > A$.

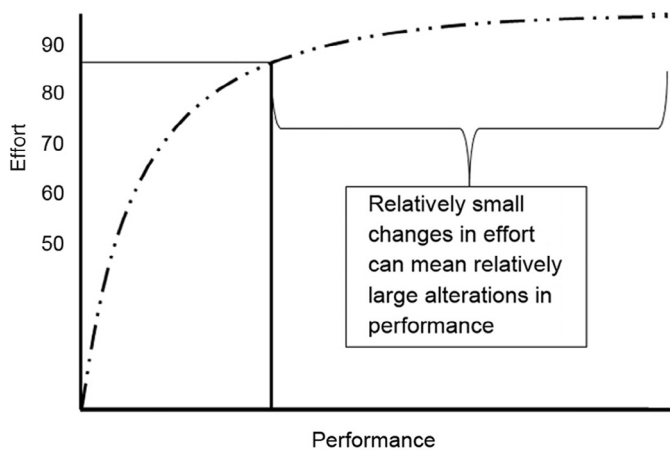


Fig. 2. Relationship between effort and performance. Based on Bannister.^{69,70}

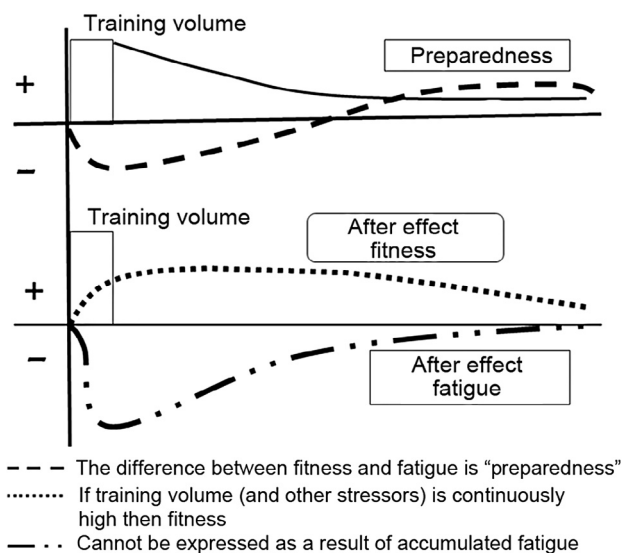


Fig. 3. Fitness-fatigue relationship. Based on Stone et al.¹ and Chiu and Barnes.⁶⁸

effort and performance.⁶⁹⁻⁷² Essentially, athletes who are encouraged to put forth high efforts and train with intent derive considerably greater performance, a factor that would transfer to both a greater training stimulus and to better competition performance. This is especially important in resistance training as maintaining higher velocities for a given load can enhance strength, velocity, and power gains even among stronger athletes compared to self-selected velocities.^{73,74} Thus, part of the commitment (and realization) on the coach's part is to instill, teach, and promote that the athlete make a maximum effort each repetition.

We must also realize that training has consequences extending far into recovery.^{25,68} Residual, or after effects, persist even after the training stimulus is reduced or terminated. The fitness-fatigue paradigm describes the interplay between the underlying mechanisms (fitness) and accumulated fatigue as a result of training (Fig. 3).

Fitness and fatigue can be conceptualized as primary after-effects of training. Fitness includes the mechanisms underlying fitness characteristics such as strength, power, and endurance. Both the volume and intensity of the training stimulus contribute to the after-effects. Accumulative fatigue is a primary cause for the inability to completely express the fitness characteristic, and thus represents the mechanisms that interfere with fitness expression. There are likely multiple fitness after-effects and multiple fatigue after-effects each associated with different underlying characteristics (e.g., strength, RFD, power, etc.). Although each specific fitness and fatigue after-effect is independent of each other, each has an additive effect. Of primary concern is the summation of fatigue after-effects (accumulative fatigue). As training progresses fitness improves and fatigue accumulates, particularly if the volume of training is increased.

The difference between fitness and fatigue is termed "preparedness" and this represents the potential to perform well. It has been shown that as the volume of training is reduced fatigue declines at a faster rate than fitness, and preparedness increases until fitness deteriorates as a result of the prolonged decrease in training volume. Thus preparedness reaches its peak at some point shortly after the training stimulus has been reduced. This is the basis of a taper and to an extent the mechanisms of planned or functional overreaching. It is also to an extent the basis of block periodization.^{7,53-55,58,75-80}

The follow-up to this article will further detail the benefits of block periodization through a discussion on how concentrated loads are sequenced and timed. Furthermore, this review will provide evidence on how periodizing the training plan through block programming can augment the strength and power characteristics required for most track and field events.

References

1. Stone M, Plisk S, Collins D. Training principles: evaluation of modes and methods of resistance training—a coaching perspective. *Sport Biomech* 2002;1:79-103.
2. Stone MH, Fleck SJ, Triplett NT, Kraemer WJ. Health- and performance-related potential of resistance training. *Sports Med* 1991;11:210-31.

3. Wernbom M, Augustsson J, Thomee R. The influence of frequency, intensity, volume and mode of strength training on whole muscle cross-sectional area in humans. *Sports Med* 2007;**37**:225–64.
4. Cormie P, McGuigan MR, Newton RU. Adaptations in athletic performance after ballistic power versus strength training. *Med Sci Sports Exerc* 2010;**42**:1582–98.
5. Cormie P, McGuigan MR, Newton RU. Influence of strength on magnitude and mechanisms of adaptation to power training. *Med Sci Sports Exerc* 2010;**42**:1566–81.
6. Cormie P, McGuigan MR, Newton RU. Changes in the eccentric phase contribute to improved stretch-shorten cycle performance after training. *Med Sci Sports Exerc* 2010;**42**:1731–44.
7. Harris GR, Stone MH, O'Bryant HS, Proulx CM, Johnson RL. Short term performance effects of high speed, high force or combined weight training. *J Strength Cond Res* 2000;**14**:14–20.
8. Stone MH, Sands WA, Stone ME. *Principles and practice of strength-power training*. Champaign, IL: Human Kinetics; 2007.
9. Epstein D. *The sports gene: inside the science of extraordinary athletic performance*. New York, NY: Penguin books; 2014.
10. Enyon N, Ruiz JR, Oliveira J, Duarte JA, Birk R, Lucia A. Genetics: a road map for future research. *J Physiol* 2011;**589**(Pt 13):3063–70.
11. Kuijper EA, Lambalk CB, Boomsma DI, van der Sluis S, Blankenstein MA, de Geus EJ, et al. Heritability of reproductive hormones in adult male twins. *Hum Reprod* 2007;**22**:2153–9.
12. Ostrander EA, Huson HJ, Ostrander GK. Genetics of athletic performance. *Annu Rev Genomics Hum Genet* 2009;**10**:407–29.
13. Wang P, Fedoruk MN, Rupert JL. Keeping pace with ACE: are ACE inhibitors and angiotensin II type 1 receptor antagonists potential doping agents? *Sports Med* 2008;**38**:1065–79.
14. Hernández D, de la Rosa A, Barragán A, Barrios Y, Salido E, Torres A, et al. The ACE/DD genotype is associated with the extent of exercise-induced left ventricular growth in endurance athletes. *J Am Coll Cardiol* 2003;**42**:527–32.
15. Ruiz JR, Arteta D, Buxens A, Artieda M, Gómez-Gallego F, Santiago C, et al. Can we identify a power-oriented polygenic profile? *J Appl Physiol* 2010;**108**:561–6.
16. Andersen LL, Aagaard P. Influence of maximal muscle strength and intrinsic muscle contractile properties on contractile rate of force development. *Eur J Appl Physiol* 2006;**96**:46–52.
17. Burke RE. The control of muscle force: motor unit recruitment and firing patterns. In: Jones NL, McCartney M, McComas AJ, editors. *Human muscle power*. Champaign, IL: Human Kinetics; 1986.p.97–106.
18. Doan BK, Newton RU, Kwon YH, Kraemer WJ. Effects of physical conditioning on intercollegiate golfer performance. *J Strength Cond Res* 2006;**20**:62–72.
19. Folland JP, Buckthorpe MW, Hannah R. Human capacity for explosive force production: neural and contractile determinants. *Scand J Med Sci Sports* 2014;**24**:894–906.
20. Girard O. Neuromuscular fatigue in tennis: mind over muscle? *ITF Coach Sport Sci Rev* 2014;**63**:7–9.
21. Bosquet L, Berryman N, Dupuy O, Mekary S, Arvisais D, Bherer L, et al. Effect of training cessation on muscular performance: a meta-analysis. *Scand J Med Sci Sports* 2013;**23**:e140–9.
22. Froböse I, Verdonck A, Duesberg F, Mucha C. Effects of various load intensities in the framework of postoperative stationary endurance training on performance deficit of the quadriceps muscle of the thigh. *Z Orthop Ihre Grenzgeb* 1993;**131**:164–7. [in German].
23. Pyne DB, Mujika I, Reilly T. Peaking for optimal performance: research limitations and future directions. *J Sports Sci* 2009;**27**:195–202.
24. Stone MH, Plisk SS, Stone ME, Schilling BK, O'Bryant HS, Pierce KC. Athletic performance development: volume load—I set versus multiple sets, training velocity and training variation. *Strength Cond J* 1998;**20**:22–31.
25. Plisk S, Stone MH. Periodization strategies. *Strength Cond J* 2003;**25**:19–37.
26. Abe T, Kojima K, Kearns C, Yohena H, Fukuda J. Whole body muscle hypertrophy from resistance training: distribution and total mass. *Br J Sports Med* 2003;**37**:543–5.
27. Wakeling JM. Motor units are recruited in a task-dependent fashion during locomotion. *J Exper Biol* 2004;**207**(Pt 22):3883–90.
28. Behm D, Reardon G, Fitzgerald J, Drinkwater E. The effect of 5, 10, and 20 repetition maximums on the recovery of voluntary and evoked contractile properties. *J Strength Cond Res* 2002;**16**:209–18.
29. Sale DG. Neural adaptations to strength training. In: Komi PV, editor. *Strength and power in sport*. London: Blackwell; 1992.p.249–65.
30. Loeb GE. Hard lessons in motor control from the mammalian spinal cord. *Trends Neurosci* 1987;**10**:108–13.
31. Wakahara T, Fukutani A, Kawakami Y, Yanai T. Nonuniform muscle hypertrophy: its relation to muscle activation in training session. *Med Sci Sports Exerc* 2013;**45**:2158–65.
32. Wakahara T, Miyamoto N, Sugisaki N, Murata K, Kanehisa H, Kawakami Y, et al. Association between regional differences in muscle activation in one session of resistance exercise and in muscle hypertrophy after resistance training. *Eur J Appl Physiol* 2012;**112**:1569–76.
33. Wells AJ, Fukuda DH, Hoffman JR, Gonzalez AM, Jajtner AR, Townsend JR, et al. Vastus lateralis exhibits non-homogenous adaptation to resistance training. *Muscle Nerve* 2014;**50**:785–93.
34. Stone MH, Sanborn K, O'Bryant HS, Hartman M, Stone ME, Proulx C, et al. Maximum strength-power-performance relationships in collegiate throwers. *J Strength Cond Res* 2003;**17**:739–45.
35. Burleson MA, O'Bryant HS, Stone MH, Collins MA, Triplett-McBride T. Effect of weight training exercise and treadmill exercise on post-exercise oxygen consumption. *Med Sci Sports Exerc* 1998;**30**:518–22.
36. McCauley GO, McBride JM, Cormie P, Hudson MB, Nuzzo JL, Quindry JC, et al. Acute hormonal and neuromuscular responses to hypertrophy, strength and power type resistance exercise. *Eur J Appl Physiol* 2009;**105**:695–704.
37. Melby CL, Scholl C, Edwards G, Bullough R. Effect of acute resistance exercise on post-exercise energy expenditure and resting metabolic rate. *J Appl Physiol* 1993;**75**:1847–53.
38. Phillips MD, Mitchell JB, Currie-Elolf LM, Yellott RC, Hubing KA. Influence of commonly employed resistance exercise protocols on circulating IL-6 and indices of insulin sensitivity. *J Strength Cond Res* 2010;**24**:1091–101.
39. Williamson DL, Kirwan JP. A single bout of concentric resistance exercises increases BMR 48 hours after exercise in healthy 59–77-year-old men. *J Gerontol* 1997;**52A**:M352–5.
40. Hornsby GA. Systematic approach to the monitoring of competitive weightlifters. Johnson City: East Tennessee State University; 2013 [Dissertation].
41. Aubry A, Hausswirth C, Louis J, Coutts AJ, LE Meur Y. Functional overreaching: the key to peak performance during the taper? *Med Sci Sports Exerc* 2014;**46**:1769–77.
42. Matveyev LP. *Fundamentals of sport training*. Moscow: FIS Publisher; 1977 1981.
43. DeWeese B, Gray HS, Sams ML, Scruggs SK, Serrano AJ. Revising the definition of periodization: merging historical principles with modern concern. *Olym Coach Magaz* 2013;**24**:5–19.
44. Stone MH. Periodization and programming for strength power sports—the short reader's digest version. Presented at the NSCA Coaches Conference, San Antonio, TX, January; 2012.
45. Dick FW. *Sports training principles*. 6th ed. Portland, OR: Kindle Publishing; 2007.
46. Mujika I. *Tapering and peaking*. Champaign, IL: Human Kinetics; 2009.
47. Mujika I, Padilla S, Pyne D, Busso T. Physiological changes associated with the pre-event taper in athletes. *Sports Med* 2004;**34**:891–927.
48. Neary JP, Bhambhani YN, McKenzie DC. Effects of different stepwise reduction taper protocols on cycling performance. *Can J Appl Physiol* 2003;**28**:576–87.
49. Issurin VB. Block periodization versus traditional training theory: a review. *J Sports Med Phys Fitness* 2008;**48**:65–75.
50. Issurin VB. New horizons for the methodology and physiology of training periodization. *Sports Med* 2010;**40**:189–206.
51. Burhus KA, Lettinich JL, Casey ML, Wilmore JH. The effects of two different types of resistance exercise on post-exercise oxygen consumption. *Med Sci Sports Exerc* 1992;**24**:S76.

52. Barzdukas A, Berning J, Bone M, Cappaert J, D'Aquisto L, Daniels J, et al. The training response of highly trained swimmers. In: Troup J, editor. *Studies by the international center for aquatic research*. Colorado Springs, CO: US Swimming Press; 1990.p.45–51.
53. Breil FA, Weber SN, Koller S, Hoppeler H, Vogt M. Block training periodization in alpine skiing: effects of 1-day HIT on $\text{VO}_{2\text{max}}$ and performance. *Eur J Appl Physiol* 2010;**109**:1077–86.
54. Garcia-Pallares J, Garcia-Fernandez M, Sanchez-Medina L, Izquierdo M. Performance changes in world-class kayakers following two different training periodization models. *Eur J Appl Physiol* 2010;**110**:99–107.
55. Issurin VB, Sahrobajko IV. Proportion of maximal voluntary strength values and adaptation peculiarities of muscle to strength exercises in men and women. Moscow: Academy of Sciences USSR. *Hum Physiol* 1985;**11**:17–22.
56. Lawton TW, Cronin JB, McGuigan MR. Strength testing and training of rowers: a review. *Sports Med* 2011;**41**:413–32.
57. Michalski RJ, Lormes W, Grunert-Fuchs M, Wodick RE, Steinnacker JM. Long term development of rowers. In: Steinnacker JM, editor. *Rowing*. Berlin: Springer; 1988.p.307–12, 19.
58. Rønnestad BR, Hansen J, Raastad T. High volume of endurance training impairs adaptation to 12 weeks of strength training in well-trained endurance athletes. *Eur J Appl Physiol* 2012;**112**:1457–66.
59. Matveyev LP, Zdornyj AP. *Fundamentals of sports training*. Moscow: Progress Publishers; 1981.
60. Verkoshansky Y. *Programming and organization of the training process*. Moscow: FIS Publisher; 1985.
61. Verkoshansky Y. Organization of the training process. *New Stud Athlet* 1998;**13**:21–31.
62. Issurin VB. Periodization training from ancient precursors to structured block models. In: Mianovic D, Sporis G, editors. *Proceedings of the 7th International Scientific Conference on Kinesiology*. Opatija, Croatia: 2014.p.374–84.
63. Stone MH, O'Bryant H, Garhammer J. A hypothetical model for strength training. *J Sports Med Phys Fitness* 1981;**21**:342–51.
64. Edington DW, Edgerton VR. *The biology of physical activity*. Boston, MA: Houghton Mifflin; 1976.
65. Fry AC, Kraemer WJ, van Borselen F, Lynch JM, Marsit JL, Roy EP, et al. Performance decrements with high-intensity resistance exercise overtraining. *Med Sci Sports Exerc* 1994;**26**:1165–73.
66. Fry AC, Schilling BK, Weiss LW, Chiu LZ. beta2-Adrenergic receptor down regulation and performance decrements during high-intensity resistance exercise overtraining. *J Appl Physiol* 2006;**101**:1664–72.
67. Moore C, Fry AC. Nonfunctional overreaching during off-season training for skill position players in collegiate American football. *J Strength Cond Res* 2007;**21**:793–800.
68. Chiu LZ, Barnes JL. The fitness-fatigue model revisited: implications for planning short- and long-term training. *Strength Cond J* 2003;**25**:42–51.
69. Banister EW. Relationship of effort and performance. Presented at *the Pre-commonwealth Games Sports Science Symposium*, Edmonton, Alberta, Canada, June; 1978.
70. Banister EW. The perception of effort: an inductive approach. *Eur J Appl Physiol* 1979;**41**:141–50.
71. Banister EW, Good P, Holman G, Hamilton CL. Modeling the training responses. In: Landers DM, editor. *The 1984 Olympic Scientific Congress proceedings. sport and elite performers*. Champaign, IL: Human Kinetics; 1986.p.7–23.
72. Banister EW. Modeling elite athletic performance. In: MacDougall JD, Wenger HA, Green HJ, editors. *Physiological testing of elite athletes*. Champaign, IL: Human Kinetics; 1998.
73. Padulo J, Mignogna P, Mignard S, Tonni F, D'Ottavio S. Effect of different pushing speeds on bench press. *Int J Sports Med* 2012;**33**:376–80.
74. Issurin V, Timofeyev V, Sharobajko I, Razumov G, Zemliakov D. Particularities of annual preparation of top-level canoe-kayak paddlers during the 1984–88 Olympic cycle. Scientific Report. Leningrad Institute of Physical Culture, 1988.
75. Mallo J. Effect of block periodization on performance in competition in a soccer team during four consecutive seasons: a case study. *Int J Perf Anal Sport* 2011;**11**:476–85.
76. Painter KB, Haff GG, Ramsey MW, McBride J, Triplett T, Sands WA, et al. Strength gains: block vs. DUP weight-training among track and field athletes. *Int J Sports Physiol Perform* 2012;**7**:161–9.
77. Bakken TA. Effects of block periodization training versus traditional periodization training in trained cross country skiers. Lillehammer: Lillehammer University College; 2013 [Master's thesis].
78. Rønnestad BR, Hansen J, Ellefsen S. Block periodization of high-intensity aerobic intervals provides superior training effects in trained cyclists. *Scand J Med Sci Sports* 2014;**24**:34–42.
79. Bartolomei S, Hoffman JR, Merni F, Stout JR. A comparison of traditional and block periodized strength training programs in trained athletes. *J Strength Cond Res* 2014;**28**:990–7.
80. Rønnestad BR, Hansen J, Thyli V, Bakken TA, Sandbakk O. 5-week block periodization increases aerobic power in elite cross-country skiers. *Scand J Med Sci Sports* 2015;doi:10.1111/sms.12418; [Epub ahead of print].